

Development of antibiotic resistance in the ocular *Pseudomonas aeruginosa* clone ST308 over twenty years

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ABSTRACT

Corneal infection caused by a bacteria *Pseudomonas aeruginosa* is common cause of ocular morbidity. Increasing antibiotic resistance by ocular *P. aeruginosa* is an emerging concern. In this study the resistome of ocular isolates of *Pseudomonas aeruginosa* clone ST308 isolated in India in 1997 (PA31, PA32, PA33, PA35 and PA37) and 2018 (PA198 and PA219) were investigated. All the isolates of ST308 had >95% nucleotide similarity. The isolates from 2018 had larger genomes, coding sequences, accessory and pan genes compared to the older isolates from 1997. The 2018 isolate PA219 was resistant to all antibiotics except polymyxin B, while the 2018 isolate PA198 was resistant to ciprofloxacin, levofloxacin, gentamicin and tobramycin. Among the isolates from 1997, five were resistant to gentamicin, tobramycin and ciprofloxacin, four were resistant to levofloxacin while two were resistant to polymyxin B. Twenty-four acquired resistance genes were present in the 2018 isolates compared to 11 in the historical isolates. All isolates contained genes encoding for aminoglycoside (*aph(6)-Id*, *aph(3')-Iib*, *aph(3'')-Ib*), beta-lactam (*blaPAO*), tetracycline (*tet(G)*), fosfomycin (*fosA*), chloramphenicol (*catB7*), sulphonamide (*sul1*), quaternary ammonium (*qacEdelta1*) and fluoroquinolone (*crpP*) resistance. Isolate PA198 possessed *aph(3')-VI*, *rmtD2*, *qnrVC1*, *blaOXA-488*, *blaPME-1*, while PA219 possessed *aadA1*, *rmtB*, *qnrVC1*, *aac(6')-Ib-cr*, *blaTEM-1B*, *blaVIM-2*, *blaPAO-1*, *mph(E)*, *mph(A)*, *msr(E)*. In both recent isolates *qnrVC1* was present in Tn3 transposon. In 219 *blaTEM-1* was carried on a transposon and *blaOXA-10* on a class 1 integron. There were no notable differences in the number of single nucleotide polymorphisms, but recent isolates carried more insertions and deletions in their genes. These findings suggest that genomes of *P. aeruginosa* ocular clonal strains with >95% nucleotide identity isolated twenty years apart had changed over time with the acquisition of resistance genes. The pattern of gene mutations also varied with more insertions and deletions in their chromosomal genes which confer resistance to antibiotics.

1. Introduction

Pseudomonas aeruginosa causes a variety of infections, including lung infections in patients with cystic fibrosis, skin infections after burns and corneal infections (microbial keratitis). The increasing prevalence of multidrug resistant (MDR) *P. aeruginosa* isolates reduces treatment options and complicates the management of these infections. Antibiotic resistance occurs mainly due to chromosomal gene mutations and possession of transferrable resistance determinants (Lister et al., 2009). MDR isolates can be clonal, particularly those associated with hospital acquired infections (Woodford et al., 2011).

Clones of *P. aeruginosa* isolated from different environments may vary. Clones may cause infection outbreaks when they enter a new environment, for example, *P. aeruginosa* isolated from water sources can also be isolated from cystic fibrosis patients (Römling et al., 1994). Diverse clonal lineages can cause disease in the lungs of cystic fibrosis patients (Römling et al., 1992). Clones of ocular isolates of *P. aeruginosa* have rarely been reported (Lomholt et al., 2001; Subedi et al., 2018). However, five multi-drug resistant *P. aeruginosa* isolates, collected in 1997 from microbial keratitis cases in India, have been reported to be clonal and of sequence type 308 (Subedi et al., 2018). ST308 is a multidrug resistant epidemic clone that has been associated with

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sporadic outbreaks of hospital infections (Jeanvoine et al., 2019). *P. aeruginosa* ST308 has been identified as a major contaminant in the hospital water plumbing networks where it was identified as a copper tolerant clone due to the presence of genomic island number 7 (Jeanvoine et al., 2019). This clone shows intraclonal diversity as it adapts to different environments and acquires genes such as those on mobile genetic elements and acquires mutations in its genome (Abdouchakour et al., 2017).

The current study investigated the genomes of MDR *P. aeruginosa* corneal ST 308 isolates collected in 2018 with those recovered from the same location in India and collected in 1997. The aim of the study was to investigate whether this clone had changed its resistome during this period.

2. Materials and methods

2.1. *P. aeruginosa* strains

All strains had been isolated from the corneas of patients with keratitis that presented at the LV Prasad Eye Institute in 1997 or 2018. These two isolates were identified from a considerably larger group of keratitis isolates with different sequence types (Khan et al., 2020a) and were compared with the seven historical isolates with the same ST308. All isolates were obtained from different patients. Sequence types were investigated using PubMLST (<https://pubmlst.org/>).

2.2. *P. aeruginosa* genomic sequencing and analysis

DNA was extracted using DNeasy Blood and Tissue Kit (Qiagen, Hilden Germany) as per the manufacturer's recommendations from two keratitis *P. aeruginosa* strains 198 and 219 isolated in India in 2018 for cases of microbial keratitis. A paired-end library was prepared using Nextera XT DNA library preparation kit (Illumina, San Diego, CA, USA). All the libraries were multiplexed on one MiSeq run. The raw reads of the sequenced genomes were analysed for their quality using FastQC version 0.11.7 (<https://www.bioinformatics.babraham.ac.uk/projects/fastqc>). Version 0.38 of the Trimmomatic (Bolger et al., 2014) was used for trimming the adapters from the reads following *de-novo* assembly using Spades v3.13.0 (Nurk et al., 2013). Genomes were annotated using Prokka v1.12 (Seemann, 2014). Resistance genes were identified using the online database Resfinder v3.1 (Centre for Genomic Epidemiology, DTU, Denmark) (Zankari et al., 2012). Mutations in the genes were detected using Snippy V2 (Seeman, 2015) with PAO1 as the reference genome. Core genome and pan genomes were analysed using Harvest Suite Parsnp v1.2 and Roary v3.11.2 respectively. Roary output 'gene.presence.absence.csv' was used to find the accessory resistance genes in the isolates. Integrations were located using Integron Finder in Galaxy v1.5.1 (Cury et al., 2016). The genes possessed by strains 198 and 219 were then compared to those from other ST308 isolates that had been previously examined (Subedi et al., 2018). Genomes of other *P. aeruginosa* isolates (Bio project PRJNA381838) were also included for comparison. Several non-clonal isolates were also added for comparison, including isolate PA17 (Accession number NZ_QDGR000000000.1), PA57 (Accession number NZ_QDGS000000000.1), PA86 (Accession number NZ_QDGD000000000.1), PA_D1 (Accession number CP012585.1) and PA_D2 (Accession number CP012578.1). These genomes were re-annotated in the current study to avoid any bias. Pairwise genome comparison of the isolates was conducted using web-based programme JSpeciesWS (<http://jspecies.ribohost.com/jspeciesws/#analyse>).

2.3. Antibiotic susceptibility

Two strains (PA198 and PA219) isolated in 2018 and five strains (PA31, PA32, PA33, PA35 and PA37) isolated in 1997 (Subedi et al., 2018) were screened for resistance to a variety of antibiotics which are commonly used to treat microbial keratitis (Willcox, 2011). This antibiotic selection (a groups of 8 antibiotics) was made based on the

different antibiotics used in different regions for the treatment of keratitis. The minimum inhibitory concentration (MIC) and minimum bactericidal concentration (MBC) of ciprofloxacin, levofloxacin, gentamicin, ceftazidime (Sigma-Aldrich, St. Louis Missouri, USA), polymyxin B (Sigma-Aldrich, Vandtårnsvej, Søborg, Denmark) tobramycin, piperacillin (Cayman Chemical Company, Ann Arbor, Michigan, USA) and imipenem (LKT Laboratories Inc, Minnesota, USA) were determined using the broth microdilution method in 96-wells plates following the Clinical and Laboratory Standard Institute CLSI guidelines (CLSI). The protocol for MIC and MBC was adapted from a previous work (Khan et al., 2020b). The protocol used for MIC and MBC was adapted from a previous publication (Khan et al., 2020b). In brief, MICs for the individual antibiotics were measured by serially diluted in Mueller-Hinton broth the antibiotics into 1×10^5 CFU/mL (colony forming unit/millilitre) of bacteria in wells of 96 well plates. The bacteria were incubated with the antibiotics for 37 °C for 18–24 h. The well with no visible growth was considered as MIC the, whereas antibiotic concentration that gave 99.99% bacterial killing when bacteria were sub-cultured onto nutrient agar was considered the MBC.

The concentrations of antibiotics tested ranged from 5120 µg/ml to 0.25 µg/ml. Susceptibilities were interpreted using the European Committee on Antimicrobial Susceptibility Testing v9 (EUCAST, 2020) and CLSI (CLSI, 2012) breakpoints for antibiotics.

3. Results

Isolates PA198 and PA219 had sequence type 308 indicating that these strains were clonally related to each other and the ST308 strains isolated in 1997 at the same hospital.

3.1. Whole genome comparison and phylogenetics of ST308

Among all the ST308 isolates, isolate PA219 had the largest genome of 7456853 base pairs (Table 1). The number of core genes and total genes were similar for all isolates, except isolate PA219 which had a larger number of total genes but a similar number of core genes to all other isolates (Table 1). The recent isolates PA198 and PA219 had acquired a large number of additional genes (Table 1).

Genes not present in all isolates were considered as accessory genes. The CARD (The comprehensive resistance genes database) output of PAO1 was compared to analyse the presence of the resistance genes in the accessory genome. In isolate PA198, 4/67 and in PA219, 4/75 resistance genes were part of accessory genome. The overlap of the presence of accessory resistance genes in all isolates is shown in Fig. 1.

Pan genome phylogeny was constructed using PAO1 (Accession number NC_002516.1) as a reference genome. The recently published *P. aeruginosa* genome sequences of ST308 strains on NCBI with Bio project number PRJNA381838 were also aligned for both core and pan genome phylogeny. These reference isolates were from non-ocular sources. In the pan genome phylogeny, all the ocular isolates clustered together in one group which included the non-ocular isolates PAO1 and PASGNDM590. The majority of non-ocular isolates were clustered into two separate groups (Fig. 2), separated from the cluster containing the ocular ST308 isolates.

When comparing the genomes of strains, isolate PA219 was used as a reference because it had the largest genome of 7456853 base pairs. Comparison of the whole genomes of the two 2018 ST308 isolates with previous isolates from cases of keratitis (Subedi et al., 2018) showed that they were >94% similar and they were >89.9% similar to other non-ocular ST308 isolates within Bio project PRJNA381838 (Table 2). Non-ST308 isolates were all less than 84% similar.

3.2. Antibiotic susceptibility of *P. aeruginosa* isolates

Antibiotic resistance was measured using the CLSI cut-off values for antibiotic susceptibility. Isolate PA219 was resistant to all antibiotics

Table 1
General features of ST308 of *P. aeruginosa* isolates.

<i>P. aeruginosa</i> isolates	Total genes	Core genes	Accessory genes	Number of coding sequences	Number of contigs	Length of genome (base pairs)	% GC	tRNA
PA31	6937	5445	1709	6619	137	7100450	66.02	69
PA32	6927	5447	1701	6611	155	7101461	66.01	69
PA33	6932	5440	1699	6609	166	7092296	66.02	69
PA35	6932	5442	1701	6611	156	7094655	66.02	69
PA37	6958	5450	1735	6645	241	7154299	65.94	69
PA198	6882	5454	1817	6727	118	7101775	66	59
PA219	7247	5451	2202	7112	163	7456853	65.9	59

Accessory genes were determined by subtracting number of core genes (4910) of the reference strain PAO1 from total number of coding sequences of each isolate.

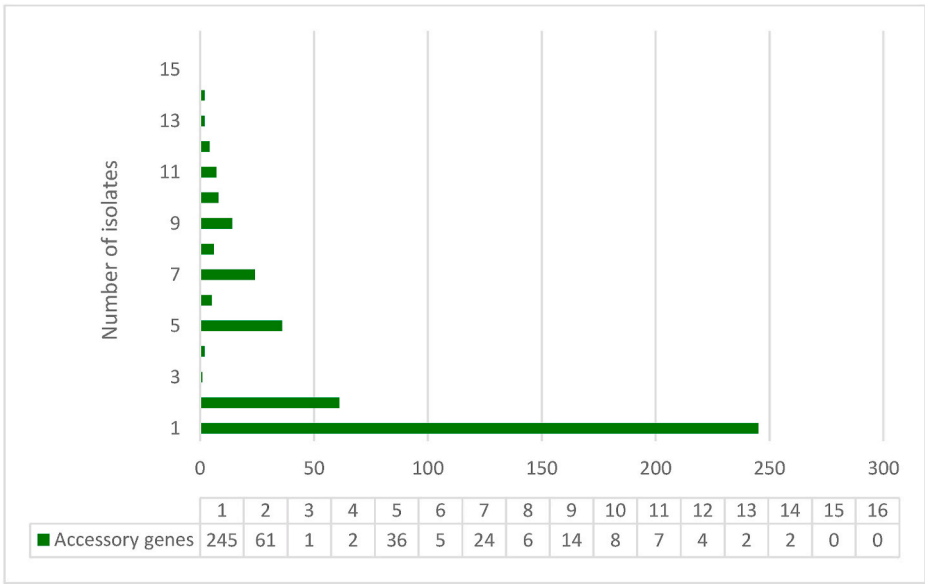


Fig. 1. Accessory resistance genes overlap in the *P. aeruginosa* isolates.

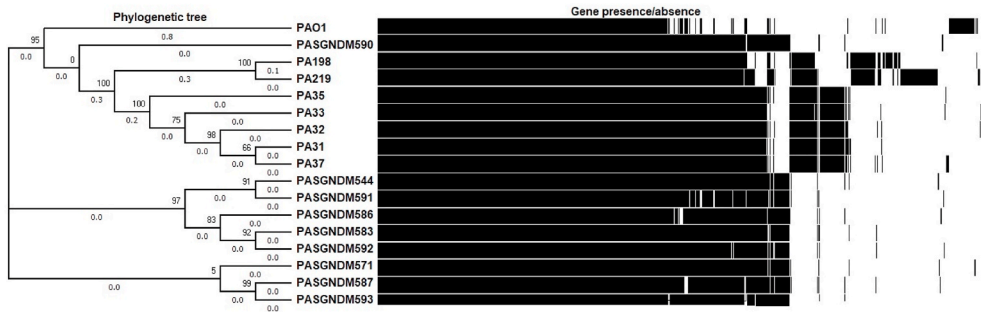


Fig. 2. Pan genome phylogeny of *P. aeruginosa* isolates using Roary. PAO1 was used as reference to build the tree. The black shading indicates the presence of genes; white indicates absence of genes. The number (ranging between 66 and 100) at each node (vertical line of the tree) represents bootstrap values. A higher bootstrap indicates higher relatedness or similarity in a tree. The number displayed on each horizontal line is the branch length, which represents an evolutionary lineage changing over time. A value 0.0 at most of the branches represents the branch length. Branch length indicates the genetic change

over time. The longer the value the more genetic change which is often measured estimating average nucleotide or protein substitution in the genome. Branch length is specified individually for each isolate in a phylogenetic tree.

other than polymyxin B (MIC = 0.5 µg/ml, MBC = 1 µg/ml). Isolate PA198 was resistant to ciprofloxacin, levofloxacin, tobramycin and gentamicin, and showed intermediate susceptibility to polymyxin B. The isolates from 1997 were all resistant to gentamicin and tobramycin and showed intermediate or definite resistance to imipenem. All five isolates from 1997 were resistant or had intermediate resistance to ciprofloxacin and four were resistant to levofloxacin. Two isolates from 1997 showed intermediate resistance to polymyxin B. Overall, the MIC and MBC values to ciprofloxacin and levofloxacin of PA198 and PA219 were higher than those recorded for the historical isolates (Supplementary Table 1).

3.3. Possession of horizontally-acquired resistance genes

In total 24 acquired resistance genes were present in the ST308 isolates of *P. aeruginosa* (Table 3). The isolates from 1997 all possessed the same 11 resistance genes. However, the isolates from 2018 had acquired additional resistance genes. Isolate PA198 carried 15 and PA219 carried 20 resistance genes. Ten resistance genes were common to all seven isolates (Table 2). These ten genes were three aminoglycoside resistance genes (*aph(6)-Id*, *aph(3'')-Ib*, *aph(3'')-Ib*), a beta-lactam resistance gene (*blaPAO*), a tetracycline resistance gene (*tet(G)*), a fosfomycin resistance gene (*fosA*), a chloramphenicol resistance gene (*catB7*) a sulphonamide resistance gene (*sul1*), and a quaternary

Table 2

Pairwise comparison of the genomes of *P. aeruginosa* isolates compared to strain PA219.

<i>P. aeruginosa</i> isolates	Isolation source	Aligned %	Aligned base pairs
ST308 isolates			
PA31	Cornea	94.09	7016166
PA32	Cornea	94.09	7015918
PA33	Cornea	94.09	7015041
PA35	Cornea	94.08	7015534
PA37	Cornea	94.09	7017651
PA198	Cornea	95.77	7141329
PASGNDM544	Endotracheal tube	90.73	6765394
PASGNDM571	Urine	90.72	6765003
PASGNDM583	Urine	90.76	6767894
PASGNDM586	Urine	89.92	6705216
PASGNDM587	Foot wound	90.14	6721892
PASGNDM590	Urine	90.69	6762596
PASGNDM591	Urine	89.93	6705872
PASGNDM592	Urine	90.74	6766496
PASGNDM593	Urine	90.42	6742830
Non-ST308 isolates			
PA17	Eye	81.34	6064371
PA57	Cystic fibrosis	77.89	5808417
PA86	Cystic fibrosis	78.15	5827469
PA_D1	Sputum	83.99	6263071
PA_D2	Sputum	83.99	6263071

PASGNDM isolates are those from Bio project PRJNA381838.

ammonium compound resistance gene (*qacEdelta1*). The recent isolates lacked one beta lactam gene (*blaOXA-50*) which was present in all the historical isolates.

Strain PA198 had acquired a 16S rRNA methylase gene (*rmtD2*) and three aminoglycoside modifying enzyme genes *aph(6)-Id*, *aph(3')-Ib*, and *aph(3'')-Ib*. Strain PA219 had acquired three different aminoglycoside resistance genes, a 16S rRNA methylase (*rmtB*), a streptomycin adenylyltransferase gene (*aadA1*) and an aminoglycoside acetyltransferase gene (*aac(6')-Ib-cr*). Strain PA219 had also acquired the plasmid-related gene *aac(6')-Ib-cr* that encodes for aminoglycoside and fluoroquinolone resistance.

One fluoroquinolone resistance gene, *crpP*, was present in all isolates from 1997 to 2018. An integron-related (Belotti et al., 2015) fluoroquinolone resistance gene *qnrVC1* was only present in isolates from 2018 (Table 3). The plasmid-related aminoglycoside and fluoroquinolone resistance gene *aac(6')-Ib-cr* was found in strain PA219 and this strain had higher MICs for ciprofloxacin and levofloxacin compared to strain PA198 and the 1997 isolates.

The metallo-beta-lactamase gene class B metallo-beta-lactamase *blaVIM-2* and a transposon (Tn2) encoded gene *blaTEM-1B* had been acquired by isolate PA219. An extended spectrum plasmid-related class A beta lactamase gene *blaPME-1* had been acquired by PA198. A class-D beta lactamase gene *blaOXA-488* had been acquired by both PA198 and PA219.

3.4. Non-synonymous mutations in the ST308 resistome

Non-synonymous mutations leading to changes in the nucleic acid sequence in the resistance genes of these *P. aeruginosa* isolates, including those related to efflux pumps, antibiotic-inactivating enzymes and drug target alterations were examined (Supplementary Table 2). The comparison of the non-synonymous mutations was made to the reference genome of strain PAO1. The number of mutations in almost all of the genes remained the same in the 1997 and 2018 isolates. However, the efflux pump gene *opmH* contained 10 SNPs in the isolates from 2018, but only 1–5 SNPs in the isolates from 1997. Similarly, *oprD* also contained 8 SNPs in the two 2018 isolates. Furthermore, non-synonymous insertions/deletions (Rietveld et al. 2004) and frame-shift mutations were found in the two isolates from 2018 (Supplementary Table 3) whereas the ST308 isolates from 1997 had no insertions/deletions or frame-shift

mutations (Subedi et al., 2018).

3.5. Mobile genetic elements

Due to the presence of a large number of acquired resistance genes in the recent isolates, the genes locations were extracted using Geneious. *QnrVC1* in the recent clones was integrated on Tn3 transposons. *QnrVC1* was accompanied by other genes on the transposon including aminoglycoside 6'-phosphotransferase (*aph(6')-Id*) and tetracycline (*tetG*) resistance genes. *BlaTEM-1B* in isolate PA219 was also integrated into the Tn3 transposon. *BlaTEM-1B* and *qnrVC1* carrying transposons were located on contig 2 and 4 respectively and so were considered to be different Tn3 transposons. In isolate PA219, a class 1 integron was also identified on contig 2 which was carried *blaOXA-10* (Supplementary Fig. 1). All the other acquired resistance genes appeared to be randomly integrated into the chromosome.

4. Discussion

This study examined whether the resistome of ST308 clones of ocular isolates of *P. aeruginosa* had changed over time. Previously, *P. aeruginosa* ST308 clones had been reported as multidrug-resistant isolates from nosocomial (Fournier et al., 2012) ocular (Subedi et al., 2018) and canine (Wang et al., 2014) infections. The ocular isolates of clone ST308 from 2018 had acquired additional resistance genes compared to those isolated in 1997 at the same hospital in India. Also, the recent 2018 isolates of ST308 had changes in the mutational patterns of the resistance genes compared to ocular isolates from 1997. The data on susceptibility of this clone of *P. aeruginosa* suggests that using polymyxin B to treat keratitis caused by this clone might be useful as most strains were susceptible, with only three strains showing intermediate susceptibility and no strain being resistant. Similarly, in a previous study that examined the antibiotic susceptibility of 24 other *P. aeruginosa* isolates from either India or Australia found that only 3/24 were resistant to polymyxin B, and so this antibiotic might be of use to treat keratitis more broadly than that caused by clone ST308.

There was >94% similarity in the genomes of the ocular ST308 isolates, indicating that these isolates were indeed very similar and can be considered to be clonally related. Of all the ocular ST308 isolates, the recent isolate PA219 had the largest genome which contained more accessory genes and coding sequences. The tRNA copy number in the recent isolates was less than those present in the historical ones. It was shown previously that the isolates with large number of acquired resistance genes by horizontal gene transfer compromise their translational machinery. In response, the tRNA copy number is retained in a genome due to the changed codon usage in the genome content (McDonald et al., 2015). All the ST308 isolates, whether ocular or non-ocular, had a similar number of core genes, perhaps indicating a conserved genome of these *P. aeruginosa* isolates (Klockgether et al., 2011; Römmling et al., 1995). However, the larger pan genome of isolate PA219 indicated that this isolate had acquired a number of genes (Klockgether et al., 2011). In the pan genome phylogeny, all ocular isolates clustered as one group and all non-ocular isolates grouped separately, which was similar to the previous findings (Subedi et al., 2018; Freschi et al., 2015).

Two different variants of 16S rRNA methylase, *rmtD2* and *rmtB*, related to aminoglycoside resistance were found in the 2018 isolates. These genes have not been reported previously in ST308 but other variants of the same genes have been identified in clones ST316 and ST235 (Tada et al., 2018). The latter clone being identified as a widespread multi-drug resistance clone. The presence of a large group of beta lactam resistance genes, specifically those acquired on mobile genetic elements is a unique finding related to ST308 in the current study. For example, a metallo-beta lactam gene including *blaTEM-1B* and a FQ resistance gene *qnrVC1* on a transposon, and *blaOXA-10* on an integron. These beta lactam resistance genes have not been reported previously in

Table 3The acquired antibiotic resistance genes in *P. aeruginosa* ST308 ocular isolates.

Antibiotic classes	Resistance genes	PA198	PA219	PA31	PA32	PA33	PA35	PA37
Aminoglycoside	<i>aph(6)-Id</i>							
	<i>aph(3')-VI</i>							
	<i>aph(3')-IIb</i>							
	<i>aph(3'')-Ib</i>							
	<i>aadA1</i>							
	<i>rmtD2</i>							
	<i>rmtB</i>							
Fluoroquinolone + Aminoglycoside	<i>aac(6')-Ib-cr</i>							
Fluoroquinolone	<i>crpP</i>							
	<i>qnrVC1</i>							
Beta lactam	<i>blaOXA-488</i>							
	<i>blaPAO</i>							
	<i>blaOXA-50</i>							
	<i>blaPAU-1</i>							
	<i>blaOXA-10</i>							
	<i>blaTEM-1B</i>							
	<i>blaVIM-2</i>							
	<i>blaPME-1</i>							
Quaternary ammoniums	<i>qacEdelta1</i>							
Sulphonamide	<i>sul1</i>							
Tetracycline	<i>tet(G)</i>							
Macrolide	<i>mph(E)</i>							
	<i>mph(A)</i>							
Macrolide, Lincosamide and Streptogramin B	<i>msr(E)</i>							
Chloramphenicol	<i>catB7</i>							
Fosfomycin	<i>fosA</i>							

Black denotes gene presence and white colour shows gene absence

strains of this clone (Subedi et al., 2018). However, the possession of the *sul1* gene in isolates of the present study was similar to a previous report (Wang et al., 2014). The possession of *blaVIM-2* and *blaTEM-1B* may have been responsible for the high MIC to piperacillin and imipenem of PA219. Previously, these genes were associated with an increased MIC of imipenem and piperacillin/tazobactam in *P. aeruginosa* isolates (Quinones-Falconi et al., 2010).

Metallo-beta lactam genes are usually found on class-1 integrons along with other antibiotic resistance determinants (Yu et al., 2006). This finding is similar to the present study but identification of a class 1 integron carrying resistance genes in the ocular *P. aeruginosa* isolates is a novel finding. The same sequence type ST308 has previously been reported to carry metallo-beta lactam *blaIMP-13* (Wang et al., 2014; Fournier et al., 2012). However, *blaTEM-1B*, *blaVIM-2* and *blaPME-1* had not been previously reported in ST308. Acquired genes within the mobile genetic elements of ST308 clones had not been identified in an earlier report (Subedi et al., 2018). Both recent isolates PA198 and PA219 had acquired genes associated with mobile genetic elements in the current study.

The presence of the plasmid-related fluoroquinolone resistance gene

qnrVC1 (Kocsis et al., 2019) and the recently reported plasmid-related gene *crpP* (Chávez-Jacobo et al., 2018) are also novel findings in the current study related to ST308 *P. aeruginosa* isolates. All isolates contained the fluoroquinolone resistance gene *crpP*, but this had not been identified as a potential plasmid related fluoroquinolone resistance gene prior to the publication of resistance genes of the 1997 isolates (Subedi et al., 2018). Usually, fluoroquinolone resistance is due to mutation in DNA gyrase and topoisomerase IV genes (Livermore, 2002). However, the very high MICs to ciprofloxacin and levofloxacin of the 2018 isolates of ST308 might be due to the acquisition of *qnrVC1*. Strain PA219 had also acquired the plasmid related fluoroquinolone resistance gene *aac(6')-Ib-cr* (Ma et al., 2009) which can confer resistance to both fluoroquinolones and aminoglycosides (Robicsek et al., 2006). Previously this gene was found to be responsible for the 16 to 128-fold higher MICs for ciprofloxacin in transconjugants of *Enterobacteriaceae* (Yang et al., 2008) and MIC of 64 µg/ml for ciprofloxacin of MDR *P. aeruginosa* non-ocular isolates (Araujo et al., 2016). These resistance elements for fluoroquinolones suggest that alternative treatments for keratitis other than fluoroquinolone monotherapy might need to be considered in the future. Furthermore, acquisition of a larger number of aminoglycoside

and beta lactam resistance genes is alarming because, where first line therapy such as monotherapy with fluoroquinolones fails, fortified antibiotics (Nixon, 2018) such as gentamicin plus cephalosporins are often prescribed.

Indels (insertion/deletion polymorphisms due to non-synonymous mutations) in the 2018 isolates of *P. aeruginosa* ST308 clones were not present in the strains isolated in 1997 (Subedi et al., 2018). Indels or frame shift mutations can potentially trigger changes in the amino acid sequence of proteins resulting in unstable protein expression or expression of non-functional proteins. Genetic mutations can lead to a change in the binding ability of ligands (McCready et al., 2019) and a potential increase in antibiotic resistance due to target alteration (Juan et al., 2005; Amin et al., 2005). Additionally, the increased number of SNPs suggests that an increase in selection pressure in the environment may have selected for these mutations. Additionally, mutations result in changes to resistance and these have been previously shown (Hirai et al., 1987; Schmitz et al., 1998; Jalal et al., 1999; Srikumar et al., 2000; Gorgani et al., 2009). Overall, large mutations were present for *opmH* and *rocS*. *OpmH* is part of an efflux pump responsible for the resistance of *P. aeruginosa* to the biocide triclosan that is often used in hospitals (Jones et al., 2000). *RocS* is a sensor kinase involved in the transduction pathway for fimbrial adhesion during biofilm formation (Kulasekara et al., 2005). No other differences were found in mutation of other genes in the ocular ST308 isolates in the present study.

The emergence of genetic variants in cystic fibrosis infection strains has been related to mutation during prolonged colonization and antibiotic administration (Bragonzi et al., 2009; Jelsbak et al., 2007). The ocular and the non-ocular isolates used for comparison in the current study were different to each other in the phylogenetic evolution. The recent ocular clones were different from the historical ones in acquisition of genes and showed different patterns of gene mutations. Furthermore, the coeval (isolated at same time) isolates PA198 and PA219 were even different from each other in the number and patterns of mutations. Genomic variations occur as a result of mutations in the genes even in the *Pseudomonas aeruginosa* isolates causing the same infection (Chung et al., 2012). The recent clones 198 and 219 with distinct features were different in their characteristics from rest of the isolates. This differentiation indicated the intraclonal diversity of isolates of this particular sequence type (Abdouchakour et al., 2017). Future research should examine other isolates from ST308 sequence types to determine how these have also changed their resistome over time.

The acquisition of resistance genes, at least in the Indian environment, may result in high levels of antibiotic resistance and this may in turn result in vision loss and blindness. Furthermore, the increasing levels of antibiotics needed to kill strains (higher MICs) suggest that the treatment of keratitis might be becoming more problematic. Studies which link genetic and phenotypic resistance to clinical outcome would help to identify the reasons for poor outcomes.

Nucleotide accession

The nucleotide sequences of recent and historical isolates are available in the GenBank under the Bio project accession number PRJNA590804 and PRJNA431326 respectively.

Authors' contributions

MW: Conceptualization of the study, manuscript review and editing.
MK: Experimental procedures, genome analysis and writing of the manuscript.

FS: Conceptualization of the study, manuscript review and editing.
SS: Donation of strains and manuscript review.

SR: Genome sequencing facilitation, manuscript review.

All authors have approved the final article.

Gene bank project number

PRJNA590804 and PRJNA431326.

Declaration of competing interest

All the authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.exer.2021.108504>.

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